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Highlights

- Backbone-based crossover combined with dedicated tabu search
- Alternating exploration in feasible and infeasible regions by constraint relaxation
- Competitive results with respect to both exact and heuristic approaches
- Hybridization of memetic algorithm with integer programming
- Proven optima for about 87 % of the problem benchmark

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A memetic algorithm for the orienteering problem with mandatory visits and exclusionary constraints

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Abstract

The Orienteering Problem with Mandatory Visits and Exclusionary Constraints (OPMVEC) is to visit a set of mandatory nodes (locations) and some optional nodes, while respecting the compatibility constraint between nodes and the maximum total time budget constraint. It is a variation of the classic orienteering problem that originates from a number of real-life applications. We present a highly effective memetic algorithm (MA) for OPMVEC combining: (i) a dedicated tabu search procedure that considers both feasible and infeasible solutions by constraint relaxation, (ii) a backbone-based crossover, and (iii) a randomized mutation procedure to prevent from premature convergence. Experiments on six classes of 340 benchmark instances from the literature demonstrate highly competitive performance of MA - it reports improved results for **104** instances compared with the existing heuristic approach, while finding matching best-known results for the remaining cases. **Additionally, MA can be used to produce a starting point for an exact solver (e.g., CPLEX), leading to an increased number of problem instances that are solved to optimality.** We further investigate the contribution of the key algorithmic elements to the success of the proposed approach.

Keywords: Heuristics; Orienteering problem; Memetic algorithm; Infeasible local search

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1 Introduction

Given a complete undirected graph $G = (V, E)$, where each node (i.e., location) $i \in V$ is associated with a non-negative score s_i and each edge $(i, j) \in E$ has a weight t_{ij} representing the travel time between two corresponding locations, the classic Orienteering Problem (OP) is to determine the shortest Hamiltonian path over a subset $S \subset V$ that maximizes the total collected score $f(S) = \sum_{j \in S} s_j$ within a restricted travel time budget T_{max} . The starting and the ending point in the path are fixed to the first and the last node in $V = \{0, 1, 2, \dots, n\}$ respectively. OP is thus a combination of node selection and finding the shortest path between the selected nodes such that each node is visited exactly once. The problem was first proposed by Tsiligrirides [26], and has also been known in the literature as the Selective Traveling Salesperson Problem [11], the Maximum Collection Problem [10] and the Bank Robber Problem [1].

The Orienteering Problem with Mandatory Visits and Exclusionary Constraints (OPMVEC) is a variant of OP, recently proposed by Palomo-Martínez et al. [22]. In OPMVEC, nodes from V are additionally grouped into two subsets of mandatory M and optional O nodes (i.e., $V = M \cup O$, $M \cap O = \emptyset$). Furthermore, some nodes are not compatible with others. For each $i \in V$, the set of incompatible nodes is denoted as C_i . A solution to OPMVEC can then be defined as a sequence $S = (v_0, v_1, \dots, v_k)$, $S \subset V$ that starts and ends at two predefined points $v_0 = 0$ and $v_k = n$ respectively, while covering all $M \subset V$ and some nodes from $O \subset V$ without violating the node compatibility and the maximum time budget constraint. More precisely, a path S is feasible in terms of the node compatibility constraints if for all $i \in S$, $S \cap C_i = \emptyset$. Given $k + 1$ nodes in S , the time budget constraint is fulfilled if $t(S) = \sum_{i=0}^{k-1} t_{v_i, v_{i+1}} \leq T_{max}$. The aim is to determine a feasible solution S^* of maximum quality, i.e., $\forall S \in \mathcal{U}, f(S^*) \geq f(S)$ where $\mathcal{U}$ is the set of all the possible feasible solutions. As OP has shown to be NP-hard in [8], this implies that OPMVEC is NP-hard.

A simple application of the problem is to determine a tourist tour that visits a set of important (must-see) sights, as well as some optional sights, within limited time while avoiding visits to very similar sights. In [24], Salazar-Aguilar et al. introduce an OPMVEC application that stems from road maintenance activities faced by the Ministry of Transport in Québec (Canada). Other applications of OPMVEC arise from transportation of hazardous materials and collection of industrial waste where certain substances must not be transported in the same vehicle to prevent from dangerous reactions [20]. Nevertheless, OPMVEC has only been studied in [22,23] to the best of our knowledge. It was first introduced in [22] along with a formulation of OPMVEC that can be optimally solved with CPLEX (12.6) for some small

instances, and a hybrid variable neighborhood search approach to deal with large instances. Recently, the authors in [23] provided five mixed integer linear formulations of OPMVEC, adopted from the Traveling Salesman Problem literature, that differ in the way the subtour elimination constraint is handled. Furthermore, different variations of OP and OPMVEC have been studied in [6,14–16,18,24,25,27].

The purpose of this work is to enrich the OPMVEC literature with a powerful memetic algorithm (MA) that integrates several complementary key components, including a dedicated tabu search procedure, a backbone-based crossover operator and a randomized mutation procedure. For an effective exploration of the search space, the tabu search procedure probabilistically selects between two evaluation functions and considers both feasible and infeasible search regions. Computational results reveal that MA clearly outperforms the exact approach to OPMVEC proposed in [23], as well as the existing heuristic from [22]. Furthermore, the combined use of the proposed MA algorithm with a mixed integer linear formulation from [23] allows CPLEX to solve more instances to optimality.

The remainder of this paper is organized as follows. Section 2 provides a mathematical formulation of OPMVEC, followed by a detailed description of MA in Section 3. A presentation of results and statistical comparisons with the existing state-of-the-art exact and heuristic methods are given in Section 4. Section 5 examines the contribution of the key MA components to the overall algorithmic performance, prior to conclusions in Section 6.

2 Mathematical formulation

Several mixed integer linear formulations have been proposed in the literature for OPMVEC [22,23]. According to the computational results reported in [23], the mathematical model (denoted as OPMVEC-GG [23]) that avoids subtours by means of a single-commodity flow formulation seems to solve the problem in the highest number of cases. Let x and y be two decision variables defined as

$$x_{ij} = \begin{cases} 1, & \text{if node } j \text{ is visited immediately after node } i \\ 0, & \text{otherwise} \end{cases} \quad (1)$$

$$y_i = \begin{cases} 1, & \text{if node } i \text{ is visited} \\ 0, & \text{otherwise} \end{cases} \quad (2)$$

Given the notation presented in the introductory section, OPMVEC-GG [23] is stated as follows:

$$\max f = \sum_{i=0}^n s_i y_i \quad (3)$$

subject to:

$$\sum_{j=1}^n x_{0j} = 1 \quad (4)$$

$$\sum_{i=0}^{n-1} x_{in} = 1 \quad (5)$$

$$y_k = 1 \quad k \in M \quad (6)$$

$$\sum_{j \in V \setminus \{n\}, i \neq j} x_{ji} = y_i \quad i \in V \setminus \{0\} \quad (7)$$

$$\sum_{j \in V \setminus \{0\}, i \neq j} x_{ij} = y_i \quad i \in V \setminus \{n\} \quad (8)$$

$$\sum_{j \in C_i} y_j + y_i \leq 1 \quad i \in V, C_i \neq \phi \quad (9)$$

$$\sum_{i \in V \setminus \{n\}} \sum_{j \in V \setminus \{0\}} t_{ij} x_{ij} \leq T_{max} \quad (10)$$

$$0 \leq g_{ij} \leq n x_{ij} \quad i \in V \setminus \{n\}, j \in V \setminus \{0\} \quad (11)$$

$$\sum_{j=0}^{n-1} g_{ji} - \sum_{j=1}^n g_{ij} = y_i \quad i \in V \setminus \{0, n\} \quad (12)$$

$$x_{ij} \in \{0, 1\} \quad i, j = 0, 1, 2, \dots, n \quad (13)$$

$$y_i \in \{0, 1\} \quad i = 0, 1, 2, \dots, n \quad (14)$$

$$g_{ij} \in Z \quad i, j = 0, 1, 2, \dots, n \quad (15)$$

The objective function (3) seeks to maximize the total collected score. Constraints (4) and (5) ensure that the path starts at node 0 and ends at node n . Constraint (6) guarantees that all the mandatory nodes are included in the path, while constraints (7) and (8) ensure flow conservation. Constraint (9) prevents conflicting nodes from being included in the path. Constraint (10) assures that the duration of the path does not exceed the time limit, while constraints (11) and (12) eliminate subtours. The remaining constraints are related to the decision variables.

3 Memetic algorithm for OPMVEC

3.1 Main scheme

Memetic algorithm (MA) [9,19,21] is a powerful framework that combines the exploration power of genetic algorithm and the exploitation strength of local search. The general architecture of our MA for OPMVEC is summarized in Algorithm 1. Starting from an initial population of feasible solutions (Section 3.2) improved with a tabu search procedure (Section 3.3), the algorithm iterates between a crossover phase and a local search improvement procedure. More precisely, the crossover consists in combining (mating) two randomly selected parent solutions to generate two offspring solutions. If the resulting offspring are not diverse, a randomized mutation procedure (Section 3.5) is applied to offspring to prevent from premature convergence. The local search procedure is a tabu search algorithm that is applied to improve the offspring, followed by a population updating procedure that updates the population with the newly created solution (Section 3.6). This process is repeated for a fixed number of generations (iterations). A more detailed description of the algorithmic components is provided in the following sections.

Algorithm 1 Memetic algorithm for OPMVEC

Require: $G = (V, E)$: a weighted undirected graph

Ensure: S^* : best solution found so far

```

1:  $P = \{S_1, S_2, \dots, S_p\} \leftarrow \text{Initial\_Population}()$  /* Section 3.2 */
2: for  $i = \{1, 2, \dots, p\}$  do
3:    $S_i \leftarrow \text{Tabu\_Search}(S_i)$  /* Section 3.3 */
4: end for
5:  $S^* \leftarrow \text{Best}(P)$  /*record the best solution found so far*/
6: while stop condition is not met do
7:   Randomly select  $S_1$  and  $S_2$  from  $P$ 
8:    $(o_1, o_2) \leftarrow \text{Crossover}(S_1, S_2)$  /* Section 3.4 */
9:   for each offspring  $o \in (o_1, o_2)$  do
10:    if  $o$  not diverse then
11:       $o \leftarrow \text{Mutate}(o)$  /* Section 3.5 */
12:    end if
13:     $S_0 \leftarrow \text{Tabu\_Search}(o)$  /* Section 3.3 */
14:    if  $f(S_0) > f(S^*)$  then
15:       $S^* \leftarrow S_0$ 
16:    end if
17:     $P \leftarrow \text{Population\_Update}(P, S_0)$  /* Section 3.6 */
18:  end for
19: end while
20: return  $S^*$ 

```

3.2 Population initialization

Each solution S from the initial population is obtained with a randomized procedure that performs in the following steps:

- (1) Initialize S by including only the predefined starting and ending nodes;
- (2) Randomly pick a mandatory node $i \notin S$ from the mandatory node set M and insert it at the best position in the current sequence S (see Function 16). This process is repeated until all the mandatory nodes are included in S ;
- (3) Use the 3-opt operator to shorten the path length of S (see Section 3.3.1);
- (4) Randomly select an optional node j from the candidate set $\{v : v \in V \setminus S, |C_v \cap S| = 0, t(S) + e_v \leq T_{max}\}$ and insert it at the best position in the current sequence S . This process is repeated until no optional node can be added without violating the maximum time limit T_{max} ;
- (5) Use the 3-opt operator to shorten the path length of S .

A node k is inserted at the best position in S according to the following function:

$$e_k = \min_{(i,j) \in S} \{t_{ik} + t_{kj} - t_{ij}\}, \quad (16)$$

where e_k is the least additional travel time and $(i, j) \in S$ is a pair of two consecutive nodes in the sequence.

Prior to insertion into the population, each newly constructed solution is further improved with a dedicated tabu search procedure that is detailed in the following section.

3.3 Tabu search

The tabu search procedure, which has the critical role in exploiting the search space, is another key component of our MA algorithm. Inspired by several node routing problems such as the Traveling Salesman Problem with Hotel Selection (TSPHS) [3], the Vehicle Routing Problem (VRP) [7] and some of its variants [4], the consideration of intermediary infeasible solutions may lead to high-quality solutions as it facilitates transition between structurally different solutions. At each phase of the local improvement, MA probabilistically selects between two evaluation functions ϕ_1 and ϕ_2 to respectively allow for violation of the travel time budget constraint and the exclusionary constraint. The extent of constraint violation is controlled with a self-adjusting penalty parameter φ . The exclusionary constraint is strictly imposed when ϕ_1 is used, while the travel time limit is never exceeded when ϕ_2 is employed.

The general framework of the proposed tabu search is summarized in Algorithm 2, while its main components are detailed in the following sections. Starting from a feasible solution S , each iteration of the tabu search consists in determining the best non-tabu solution (with a maximum ϕ value) from the union of the three neighborhoods of S obtained with three simple move operators (ADD, DROP, SWAP). Due to its computational complexity, a 3-opt move operator is only applied once in u_c iterations [3], to shorten the length of the Hamiltonian path of S . The φ parameter is reduced (increased) by a factor of 2 if feasible (infeasible) solutions have been achieved in u_p consecutive iterations. Finally, the best-found solution S^* is updated with S if S is feasible, and if its quality in terms of the total collected score is better than that of S^* . If an improvement with respect to S' or S^* was not achieved during L consecutive iterations, the best found solution S^* is returned as the final output of the tabu search procedure.

3.3.1 Move operators and evaluation functions

The four move operators used by the tabu search procedure are defined as follows:

- ADD: insert an optional node that is not included in the current solution S at the best position in S ;
- SWAP: exchange an optional node that is not in S with an optional node in S ;
- DROP: delete an optional node from the current solution;
- 3-OPT: this operator is used to shorten the current path. It involves deleting 3 edges in one route, reconnecting this route in all other possible ways, and then selecting the best path among the newly formed ones. This move operator has been previously applied in [3].

At each iteration, the proposed tabu search selects a non-prohibited solution (Section 3.3.2) from the combined ADD, SWAP and DROP neighborhoods with the aim to maximize the evaluation function value ϕ . At each phase of local improvement, the algorithm probabilistically selects between two evaluation functions ϕ_1 and ϕ_2 that are defined as:

$$\phi_1(S) = f(S) - \varphi * EX_1(S) \quad (17)$$

$$\phi_2(S) = f(S) - \varphi * EX_2(S) \quad (18)$$

where $EX_1(S) = \max\{t(S) - T_{max}, 0\}$ is the extent of the time budget violation, $EX_2(S)$ is the total score of incompatible nodes in S , and φ is

Algorithm 2 Tabu search with constraint violation

Require: $G = (V, E)$: a weighted undirected graph; S_0 : an initial feasible solution; L : search depth ; u_c : frequency of using 3-opt; u_p : frequency of updating the penalty parameter φ

Ensure: S^* : best feasible solution found so far

```

1:  $Iter \leftarrow 0$  /*iteration counter*/
2:  $NI \leftarrow 0$  /*number of consecutive iterations without improvement of  $S^*$  or  $S'^*$ */
3:  $S' \leftarrow S_0$  /*solution of the best objective value*/
4:  $S^* \leftarrow S_0$  /*the best feasible solution*/
5:  $S \leftarrow S_0$ 
6: Initialize tabu list and tabu tenure
7: while  $NI < L$  do
8:    $S \leftarrow Select\_Best\_Unprohibited\_Solution(S)$  /*selects the best non-tabu solution from the combined ADD, SWAP and DROP neighborhoods (see Section 3.3.1)*/
9:   Update tabu list /*Section 3.3.2*/
10:   $NI \leftarrow NI + 1$ ,  $Iter \leftarrow Iter + 1$ 
11:  if  $Iter \bmod u_c = 0$  then
12:    Use 3-opt to shorten the path length of  $S$ 
13:  end if
14:  if  $S$  is a feasible solution and  $f(S) > f(S^*)$  then
15:     $S^* \leftarrow S$ 
16:     $NI \leftarrow 0$ 
17:  end if
18:  if  $\phi(S) > \phi(S')$  then
19:     $S' \leftarrow S$ 
20:     $NI \leftarrow 0$ 
21:  end if
22:  if  $Iter \bmod u_p = 0$  then
23:    Update penalty parameter  $\varphi$ 
24:  end if
25: end while
26: return  $S^*$ 

```

the self-adjusting penalty coefficient that controls the degree of infeasibility introduced into the search.

It is notable that the same penalty coefficient φ is used in both evaluation functions as the tabu search relies only on a single evaluation function throughout the entire local improvement phase. Initially set to 1, φ is dynamically adjusted during the search according to the search history. More precisely, $\varphi = \varphi/2$ if all the u_p consecutively visited solutions are feasible (u_p is a parameter). If all the u_p consecutively visited solutions are infeasible, $\varphi = 2\varphi$. The motivation behind this dynamic strategy is to keep a balanced

search between the feasible and the infeasible search regions. If the search is too confined to the feasible search region, φ will be reduced to encourage the search to move into the infeasible region. On the other hand, if too many infeasible solutions are visited, φ will be increased to force the search back to the feasible region.

Inspired by a selection mechanism by Martí et al. [17], the probability of selecting one evaluation function over another is proportional to the number of high-quality solutions reached when considering the given evaluation function. We regard a solution to be of high quality if it is introduced into the population pool during the population update phase. More precisely, the probability of using ϕ_i is computed as:

$$P(i) = \frac{50 + q_i}{100 + q_1 + q_2},$$

where q_i counts the number of times an offspring has been inserted into the population after local improvement using ϕ_i . It is easy to observe that the higher the value of q_i , the more probable it is to use ϕ_i in the following improvement phase.

3.3.2 Tabu list management and aspiration criterion

To prevent from short-term cycling, a node that is removed from the current solution S (by SWAP or DROP) or inserted into S (by ADD or SWAP) cannot be displaced for the next tl iterations (tabu tenure). The tabu tenure tl is dynamically determined as $tl = \alpha * |O| + \text{random}(10)$, where $\text{random}(10)$ takes a random number in $\{1, \dots, 10\}$, $|O|$ is the size of the optional node set and α is the tabu tenure management factor.

Along with this tabu rule, a simple aspiration criterion is used to override the tabu restrictions if the move leads to an improved feasible solution or to a solution of a higher evaluation function value ϕ .

3.4 Crossover operator

At each generation of our memetic algorithm, a crossover operator is applied to create two new offspring individuals by recombining two randomly selected parent solutions from the population. It has been commonly recognized that the success of a memetic algorithm crucially depends on the recombination operator [9], which not only should generate diversified solutions but should also transfer meaningful components from parents to offspring. Inspired by

[28,29], we adopt a backbone-based crossover operator, where the “backbone” of S_1 and S_2 refers to the subset of nodes shared in both parents, i.e., $S_1 \cap S_2$.

Given two parents S_1 and S_2 , offspring o_1 and o_2 are constructed in the following steps:

- (1) Identify the subset of common nodes in S_1 and S_2 , i.e., $S_0 = \{v : v \in S_1, v \in S_2\}$.
- (2) Copy all the nodes from S_0 to o_1 and o_2 by keeping their order in S_1 and S_2 respectively.
- (3) If $t(o_1) < T_{max}$, select a node k with the largest value $\frac{s_k}{e_k}$ from the candidate set $\{v : v \in V \setminus o_1, |C_v \cap o_1| = 0, t(o_1) + e_v \leq T_{max}\}$ and insert it at the best position in o_1 . Repeat the process until no new node can be added without violating the maximum time limit T_{max} . o_2 is obtained in the same way.

3.5 Mutation operator

If all the nodes from the backbone $S_1 \cap S_2$ have the same order in both parents, each newly generated offspring o is mutated with a randomized mutation procedure that consists in applying η random *AddMove* or *SwapMove* moves to o , where η is a parameter called the mutation strength. These mutation moves are defined as:

AddMove: randomly select a node from the set $\{v : v \in V \setminus o\}$ and insert it at the best position in o .

SwapMove: swap a randomly selected node from the set $\{v : v \in V \setminus o\}$ with a randomly selected optional node in o .

Note that the application of *AddMove* and *SwapMove* moves may result in infeasibility in terms of the compatibility and/or the travel time limit constraint. The compatibility conflicts are eliminated by randomly selecting a node from the candidate set $\{v : v \in o \setminus M, |C_v \cap o| > 0\}$ and removing it from o . This procedure is repeated until all the nodes in o are compatible. The time budget violation is eliminated by randomly deleting an optional node from o , until a feasible solution in terms of the maximum time limit has been reached.

3.6 Population updating rule

Following the creation of an offspring o and its further improvement with the tabu search procedure, o replaces the worst solution $S_w, \forall S \in P, f(S) \geq f(S_w)$

from the population if $f(o) > S_w$. One potential risk of this classic replacement strategy is premature convergence, since o is introduced in P regardless of its distance (i.e., similarity) to other individuals in the population. The proposed MA avoids premature convergence and ensures a diversified offspring through application of the randomized mutation procedure described in Section 3.5. Thus our memetic algorithm is not really sensitive to the population replacement rule.

4 Computational experiments

The purpose of this section is to evaluate the performance of the proposed MA by means of extensive comparisons with the existing **exact and heuristic approaches** from the literature.

4.1 Benchmark and experimental protocol

For performance assessments, we use 340 benchmark instances from [22], grouped into six classes. These are based on 15 instances introduced by Fischetti et al. [5], which are commonly used to test OP and VRP approaches. For each Class 1-6, Table 1 indicates the number of instances, the mandatory node ratio and the incompatibility level. The incompatibility level is deemed low if more than half of the nodes in an instance are non-conflicting, and is high otherwise. The number of nodes ranges from 21 to 262.

Table 1
Benchmark instances

Class	Number of instances	Mandatory node ratio(%)	Incompatibility level
1	62	10	high
2	55	20	high
3	53	30	high
4	62	10	low
5	55	20	low
6	53	30	low

Our MA algorithm is coded in C++ and compiled with a g++ compiler using option ‘-O3’. All experiments are performed on a computer with an Intel Xeon(R) CPU E5-2695 v4 processor (2.1 GHz CPU and 2 GB RAM) running under Linux operating system. We perform **30** independent runs of MA per test case, with the termination criterion set to 50 generations per execution. The setting of parameters is given in Table 2. All the parameters ($P, L, \alpha, u_c, u_p, \eta$) were tuned with Iterated F-race (IFR) [2], an automatic configuration method that is part of the IRACE package [12]. The tuning was performed on a sample of 24 representative instances of varying sizes, selected from each of the six classes. For each parameter, IFR requires a limited set of values as input to choose from (column “Considered values”). The total time budget for IRACE

was set to 5000 executions of MA with 50 generations per MA execution. The suggested setting of parameters by IFR is presented in Column “Final value”.

Table 2
Parameter tuning results

Parameter	Section	Description	Considered values	Final value
P	3.2	Population size	{5,10,20}	5
L	3.3	Search depth during tabu search	{100,150,200}	200
α	3.3	Tabu tenure management factor	{0.05,0.1,0.15}	0.1
u_c	3.3	Frequency of using 3-opt	{2,5,10}	5
u_p	3.3	Frequency of updating φ	{5,10,15}	10
η	3.5	Mutation strength	{2,5,10}	5

To evaluate the performance of MA, we first present comparisons with the existing heuristic - a hybrid Variable Neighborhood Search (VNS) algorithm proposed in [22] which, to the best of our knowledge, is the only heuristic in the OPMVEC literature. The results reported by VNS were obtained after 10 executions per instance on a 2.1 GHz Intel Xeon(R) CPU E5-2620 v2 with Ubuntu 14.04 operating system. For a relatively fair comparison between MA and VNS, we use a procedure described in [4] to scale the CPU times reported for VNS in the corresponding paper. We further provide comparisons with the results obtained with five optimization models from [23], executed under CPLEX within a time limit of 1h. Finally, we show results of a hybrid between MA and an existing Integer Programming (IP) model where MA is used to obtain a starting solution for CPLEX.

4.2 Computational results and comparisons with the existing heuristic approach

This section outlines and statistically compares the performance of MA with the existing heuristic from the literature. To compare the performances of the two algorithms in terms of the average solution quality across all the problem instances from a given benchmark class, we normalize the obtained average objective values according to the following relation:

$$y = 100 * (f_{BK} - f_{AVG}) / f_{BK},$$

where f_{BK} is the best-known objective value and f_{AVG} is the average objective value. For each algorithm Alg , we then compute a sample set $Y_{Alg} = \{y_1, y_2, \dots, y_n\}$ consisting of the normalized average objective values for each of the n instances from the given benchmark class. Intuitively, values close to zero indicate the best performance, while the performance is worse as the values increase to 100.

Figure 1 provides boxplot graphs to compare the distributions of the normalized values in Y_{MA} and Y_{VNS} . To determine whether there exists a significant difference in the observed performances between the two algorithms on each set of instances, we perform pairwise Wilcoxon-signed rank tests on

the sample sets Y_{MA} and Y_{VNS} . The corresponding p -values are given in Figure 1. We observe a significant difference in performance in favor of MA on all but the third benchmark class.

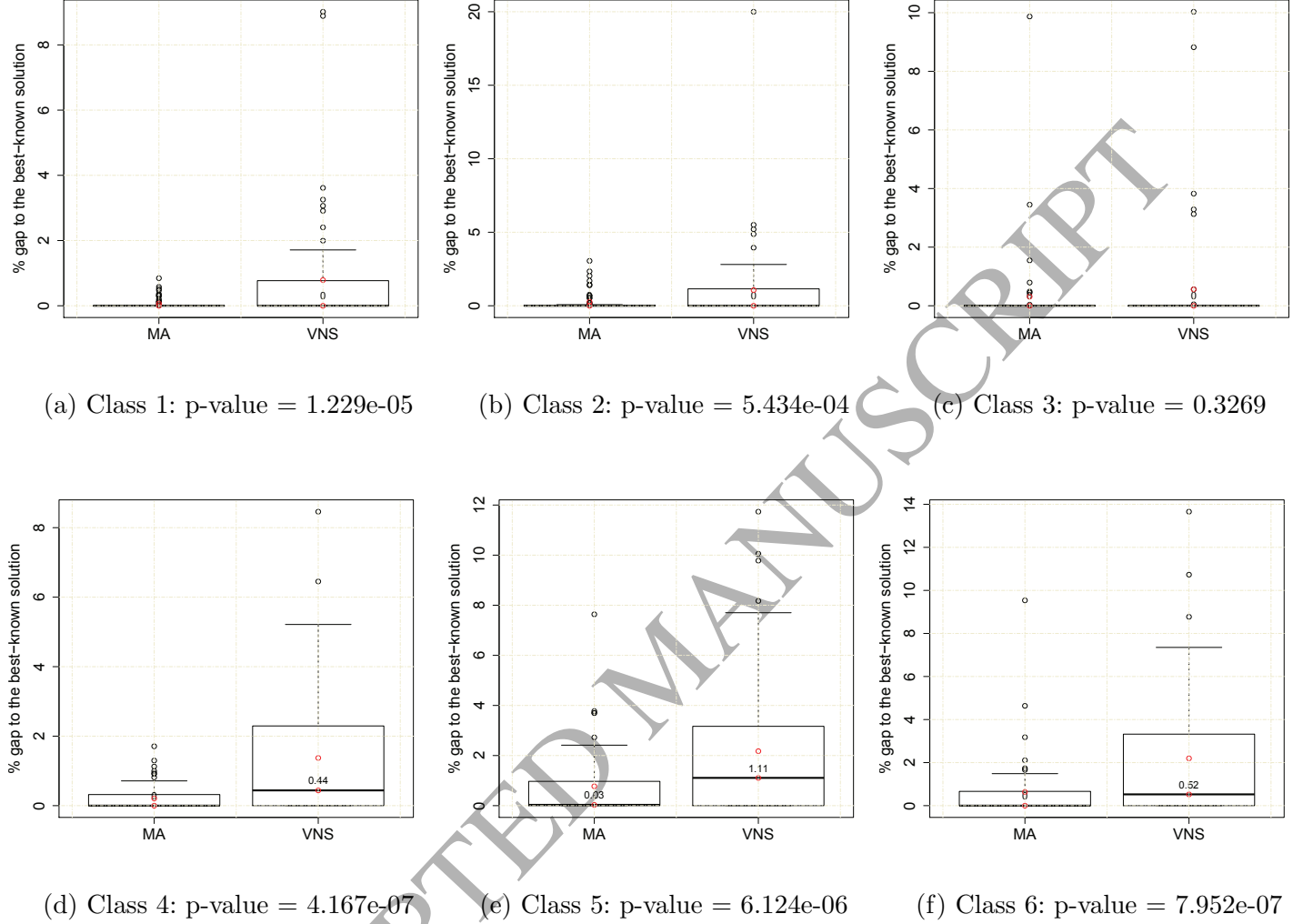


Fig. 1. Boxplots of the normalized average objective values for each individual benchmark Class 1-6.

Detailed comparisons between MA and VNS per instance are presented in the Appendix section. In a nutshell, we observe from Table 9-14 (see Appendix) that MA is able to improve the best performance reported by VNS for 104 out of the 340 instances, and reach the best result reported with VNS for the remaining 236 instances. In terms of the average objective values, MA exhibits a better average performance on 148 out of the 340 instances and is outperformed by VNS only on 7 instances. Table 3 summarizes the results reported with the two algorithms on Class 1-6. For each algorithm and benchmark class, the first row shows the number of cases for which the best-known solution was reached or improved, the second row indicates the average

gap between the best-known and the best reported solution, while the last row provides the average computing time in seconds required for each algorithm.

Table 3

Summary of results

	Class 1		Class 2		Class 3		Class 4		Class 5		Class 6	
	MA	VNS	MA	VNS	MA	VNS	MA	VNS	MA	VNS	MA	VNS
Best-known solutions	62/62	49/62	55/55	43/55	53/53	49/53	62/62	37/62	55/55	31/55	53/53	27/53
Avg. % gap to best-known sol.	0.00	0.33	0.00	0.64	0.00	0.30	0.00	0.77	0.00	1.06	0.00	1.35
Average time(s)	7.4	12.1	9.7	5.2	9.8	1.4	21.1	40.5	34.6	32.2	46.9	23.9

While the computational results are clearly in favor of MA, the performance of VNS is controlled only by a single parameter (verses 6 parameters for MA), which brings us to the trade-off between simplicity and performance. As demonstrated from a parameter sensitivity analysis in the following section, the performance of MA is mainly influenced by parameters L and u_c that control the duration and the intensity of the local search phase, leading to a trade-off between solution quality and computing time.

4.3 Comparasions with exact approaches

To further confirm the effectiveness of the proposed MA algorithm, this section compares the results of MA with those obtained with the five models (OPMVEC-DL, OPMVEC-GG, OPMVEC-W, OPMVEC-DFJ, and OPMVEC-C) from [23] that are executed by CPLEX 12.6 on a 2.10 GHz Intel Xeon(R) CPU E52620 v2 under Ubuntu 14.04 operating system.

Table 4

Comparison among the optimal solutions reported by the five models in [23] and the ones obtained with MA.

Class	OPMVEC-DL	OPMVEC-GG	OPMVEC-W	OPMVEC-DFJ	OPMVEC-C	MA
1	39/62	44/62	36/62	43/62	39/62	53/62
2	35/55	41/55	33/55	49/55	41/55	51/55
3	38/53	40/53	31/53	52/53	47/53	52/53
4	34/62	44/62	32/62	31/62	32/62	48/62
5	29/55	42/55	28/55	34/55	34/55	45/55
6	22/53	37/53	28/53	34/53	36/53	42/53
Percentage	57.94%	72.94%	55.29%	71.47%	67.35%	85.59%
Total	197/340	248/340	188/340	243/340	229/340	291/340

Table 4 shows the number of instances solved to optimality for each of the five models. Furthermore, we report the number of cases for which MA was able to attain a proven optimum. It is worth mentioning that the five models can solve to optimality a total of 277 out of the 340 instances. A combined use of MA and the OPMVEC-GG model, where MA is used to obtain a starting point for CPLEX, is able to solve for the first time 22 additional instances and approximately 87% of the problem benchmark (see next Section). From Table 4, we observe that MA outperforms each model on all the instance classes and

reaches a proven optimum for 85.59% of the instances (i.e., for a total of 291 instances).

Table 5
MA vs. five models in [23]

	MA on Class 1			MA on Class 2			MA on Class 3			MA on Class 4			MA on Class 5			MA on Class 6		
	Better	Equal	Worse	Better	Equal	Worse	Better	Equal	Worse	Better	Equal	Worse	Better	Equal	Worse	Better	Equal	Worse
OPMVEC-DL	23	39	0	20	35	0	15	38	0	28	34	0	26	29	0	31	22	0
OPMVEC-GG	18	44	0	14	41	0	13	40	0	18	44	0	13	42	0	14	39	0
OPMVEC-W	26	36	0	22	33	0	22	31	0	30	32	0	27	28	0	25	28	0
OPMVEC-DFJ	20	42	0	6	49	0	1	52	0	33	29	0	21	34	0	19	34	0
OPMVEC-C	24	28	0	14	41	0	6	47	0	30	32	0	21	34	0	18	35	0

Table 5 indicates the number of times MA reports a better, equal, or worse result compared to each model and each instance class. In each case, we notice that MA always obtains a result that is as least as good as that obtained with the given exact approach. Furthermore, it outperforms each model for a significant number of cases.

4.4 Hybridization between memetic algorithm and integer programming

This section investigates the benefit of a combined use of the proposed MA with Integer Programming (IP), where MA is used to produce a starting point for CPLEX 12.6. For this purpose, the IP model is the existing model (denoted OPMVEC-GG) from [23], detailed in Section 2. For each class of benchmark instances, Table 6 shows the number of instances that were solved to optimality with OPMVEC-GG alone, as well as with the MA+OPMVEC-GG hybrid. For a relatively fair comparison, the maximum computation time allowed to solve the model for each instance is set to one hour. Detailed results are provided in Table 9-14 of the Appendix.

Table 6
Computational results obtained with OPMVEC-GG and the proposed MA+OPMVEC-GG hybrid.

Class	Number of instances	OPMVEC-GG		MA+OPMVEC-GG	
		Optimal solutions	Time (s)	Optimal solutions	Time (s)
1	62	44	74140.6	53	42637.4
2	55	41	53649.0	52	18052.4
3	53	40	49951.3	52	4368.9
4	62	44	72496.2	50	58401.1
5	55	42	55295.1	47	39539.2
6	53	37	61918.5	45	36633.3
Total	340	248	367450.7	299	199632.3

We can conclude from Table 6 that combining MA with the given exact approach leads to a significantly higher number of solved instances than with the exact approach alone. More precisely, the MA+OPMVEC-GG hybrid was

able to solve 299 out of the 340 instances (approximately 87%), compared to 248 instances (approximately 72%) solved with OPMVEC-GG. Furthermore, MA+OPMVEC-GG requires significantly less time than OPMVEC-GG to reach the reported results, which constitutes a strong argument for the combined use of the proposed MA with an exact approach.

5 Analyses and discussions

We now turn our attention to an analysis of several major ingredients of the proposed algorithm: the consideration of both feasible and infeasible solutions, the adaptive selection strategy of evaluation functions during tabu search, and the mutation operator. **We further compare several versions of our algorithm to pinpoint the most important feature of the proposed MA.** Finally, we provide a sensitivity analysis of the MA parameters.

5.1 Impact of exploring both feasible and infeasible search space

Table 7

Comparison of results obtained with MA and its weakened version MA_w that only explores the feasible search space.

Instance	Class	N	MA			MA_w		
			f_{best}	f_{avg}	$time_{avg}(s)$	f_{best}	f_{avg}	$time_{avg}(s)$
att48C	1	48	9	9.0	0.2	9	9.0	0.0
eil76E	1	76	367	367.0	0.8	367	365.8	0.1
eil101A	1	101	570	567.3	2.5	570	565.9	2.1
cmt151C	1	151	435	435.0	4.2	430	420.5	0.2
op21D	2	21	75	75.0	0.0	75	75.0	0.0
eil51C	2	51	168	168.0	0.3	168	168.0	0.0
cmt121B	2	121	305	304.4	4.7	305	302.8	3.2
cmt151A	2	151	462	458.8	6.1	460	455.5	3.2
att48A	3	48	4	4.0	0.2	4	4.0	0.0
eil76C	3	76	142	137.1	0.9	139	125.0	0.0
cmt200B	3	200	391	387.9	35.5	391	358.7	0.6
gil262C	3	262	1369	1368.5	84.7	1366	1365.1	69.5
att48B	4	48	29	29.0	2.4	29	28.6	0.1
eil51A	4	51	328	328.0	0.7	328	328.0	0.3
cmt200A	4	200	723	713.6	22.8	723	694.2	0.4
gil262E	4	262	4413	4371.7	106.9	4273	4197.2	4.6
eil51B	5	51	236	236.0	0.9	236	233.1	0.2
eil76E	5	76	411	408.2	2.1	411	407.3	0.1
eil101E	5	101	521	506.8	5.7	492	484.3	0.3
gil262B	5	262	3618	3484.1	149.2	3511	3424.2	8.2
op32A	6	32	115	115.0	0.1	115	115.0	0.1
att48B	6	48	19	19.0	1.6	19	19.0	0.3
cmt121D	6	121	618	617.8	33.1	618	618.0	2.2
gil262C	6	262	4070	3940.6	337.6	4053	3893.0	22.3

As explained in Section 3.3, our tabu search procedure considers both feasible and infeasible regions by relaxing the time limit constraint or the node compatibility constraint. To evaluate the merit of this feature on a selection of 24 representative instances, we compare our algorithm with a weakened version of MA (denoted as MA_w) that only explores the feasible search space. The experimental results, obtained under the same condition as specified in Section 4.1, are shown in Table 7.

It can be observed that MA outperforms MA_w in terms of both the best (p-value = 0.011616) and the average (p-value = 0.0003515) performance.

5.2 Impact of the joint use of two evaluation functions

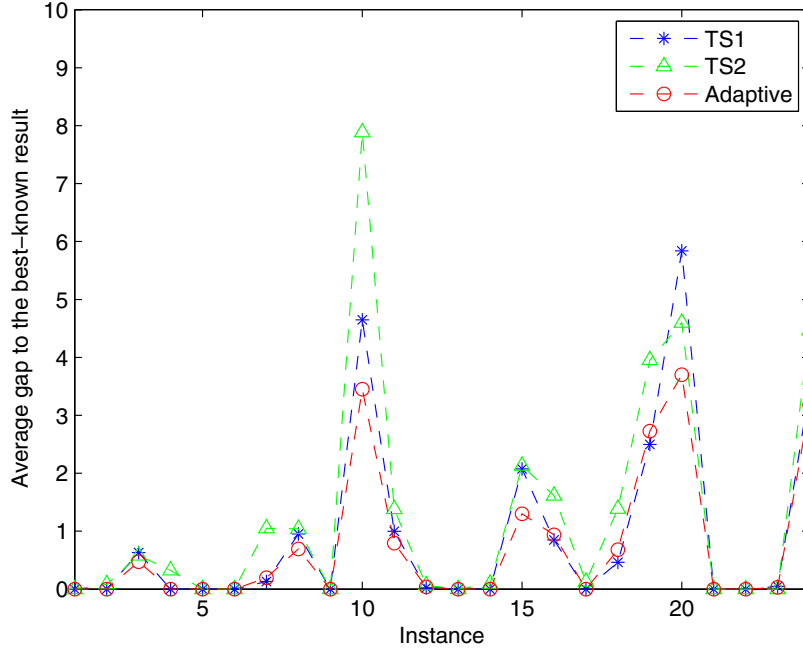


Fig. 2. Percentage gap between the best-known and the average result reported with MA, TS1 and TS2 on a selection of 24 instances.

As described in Section 3.3, MA probabilistically selects between two evaluation functions prior to offspring improvement. To investigate the benefit of this feature, we compare our algorithm with two weakened versions of MA that make use of a single evaluation function, denoted as TS1 (version that only uses the evaluation function from Eq. 17) and TS2 (version that only uses the evaluation function from Eq. 18). Figure 2 shows the percentage gap between the best-known solution and the average result reported with the three algorithms on a selection of 24 instances. These results clearly highlight the contribution of the combined use of the two evaluation functions.

5.3 Influence of mutation operator

Mutation operator is another important ingredient of the proposed MA algorithm. As explained in Section 3.5, a randomized mutation procedure is used to prevent from premature convergence by ensuring offspring diversity.

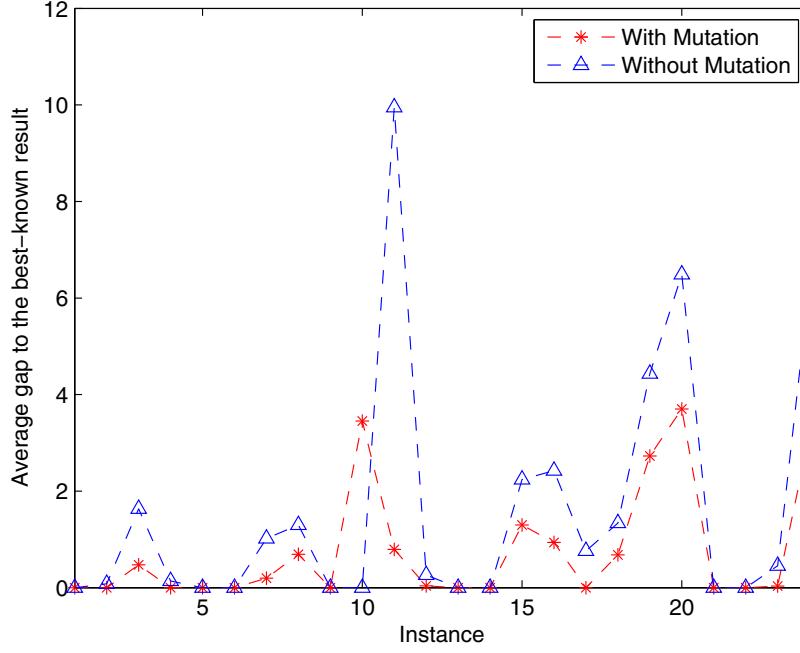


Fig. 3. Percentage gap between the best-known and the average result obtained with two MA versions with and without the mutation operator.

To evaluate the effect of the mutation operator to the MA performance, Figure 3 compares two MA versions with and without the mutation operator on the subset of 24 selected instances. For each instance, the y -axis indicates the gap between the best-known result and the average result obtained by each algorithm over 30 independent runs. The figure clearly highlights the benefit of the mutation operator.

5.4 Identification of the most crucial MA feature

The proposed MA integrates several important elements that jointly contribute to its high efficiency, including: (i) adaptive alternation between exploitation in feasible and infeasible regions through constraint relaxation, (ii) joint use of two evaluation functions, (iii) backbone-based crossover, and (iv) randomized mutation procedure. To identify the most important feature of our MA, we evaluate several MA variants that exclude different elements: (i) a variant that only exploits the feasible search region, (ii) a variant that does not make use of the mutation operator, (iii) a variant that only makes use of the first evaluation function (Eq. 17), (iv) a variant that only considers the second evaluation function (Eq. 18), and (v) a variant without the crossover operator. To implement the MA variant without the crossover operator, we simply select and mutate two parents from the population. Aside from the

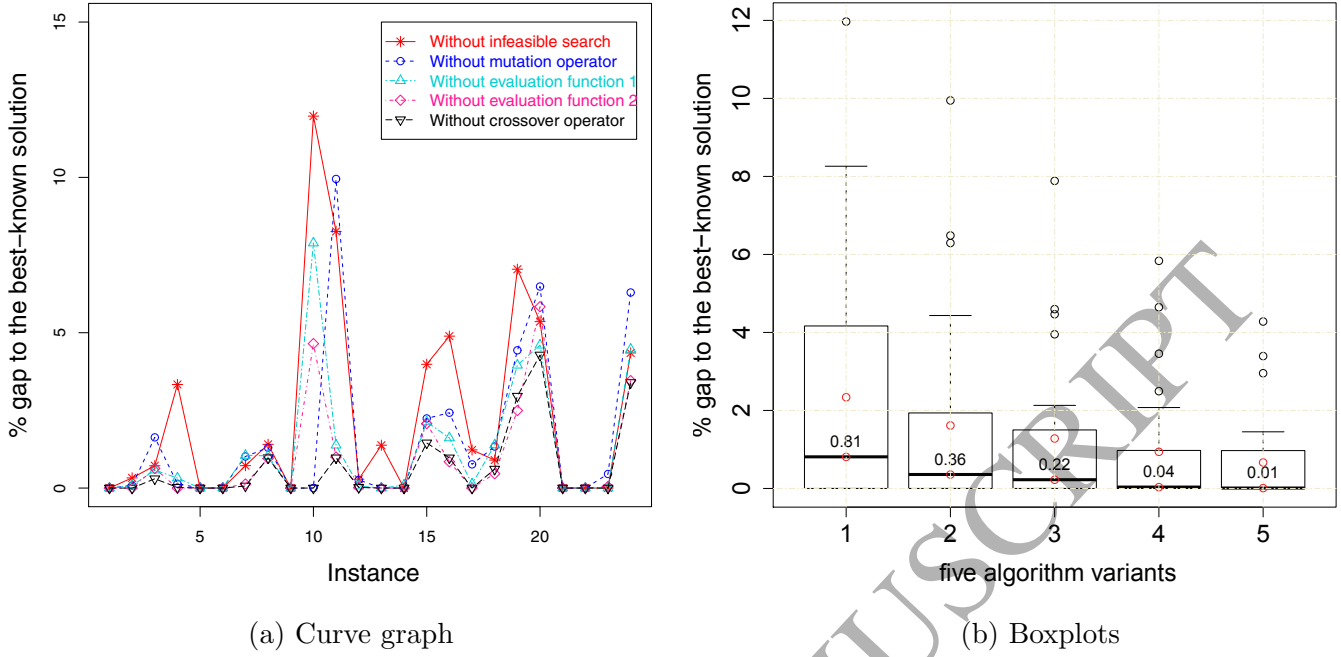


Fig. 4. Computational results of the five MA variants on a selection of 24 instances.

feature being tested, the other elements are kept unchanged. We evaluate each MA version on the selection of 24 instances with 30 independent runs per instance.

For each MA version, Figure 4(a) shows the percentage gap between the best-known solution and the average result reported with the given algorithm on the 24 used instances. According to these results, the variant that limits the search to the feasible region only reports the worst performance. This highlights the benefit of considering solutions that are infeasible up to a certain degree, as such transitions may be required to reach high quality feasible solutions. Figure 4(b) further shows box and whisker plots that indicate the distribution and the range of the normalized average objective values for each MA variant. We observe a visible difference in the distributions of results, that further confirms the weakness of the first variant (without infeasible search).

5.5 Sensitivity analysis of important parameters

We analyze the sensitivity of the six MA parameters: the population size (P), the search depth during tabu search (L), the tabu tenure management factor (α), the frequency of using 3-opt (u_c), the frequency of updating the penalty coefficient φ (u_p) and the mutation strength (η). To evaluate the sensitivity of each parameter, we test the considered values for each parameter from Table

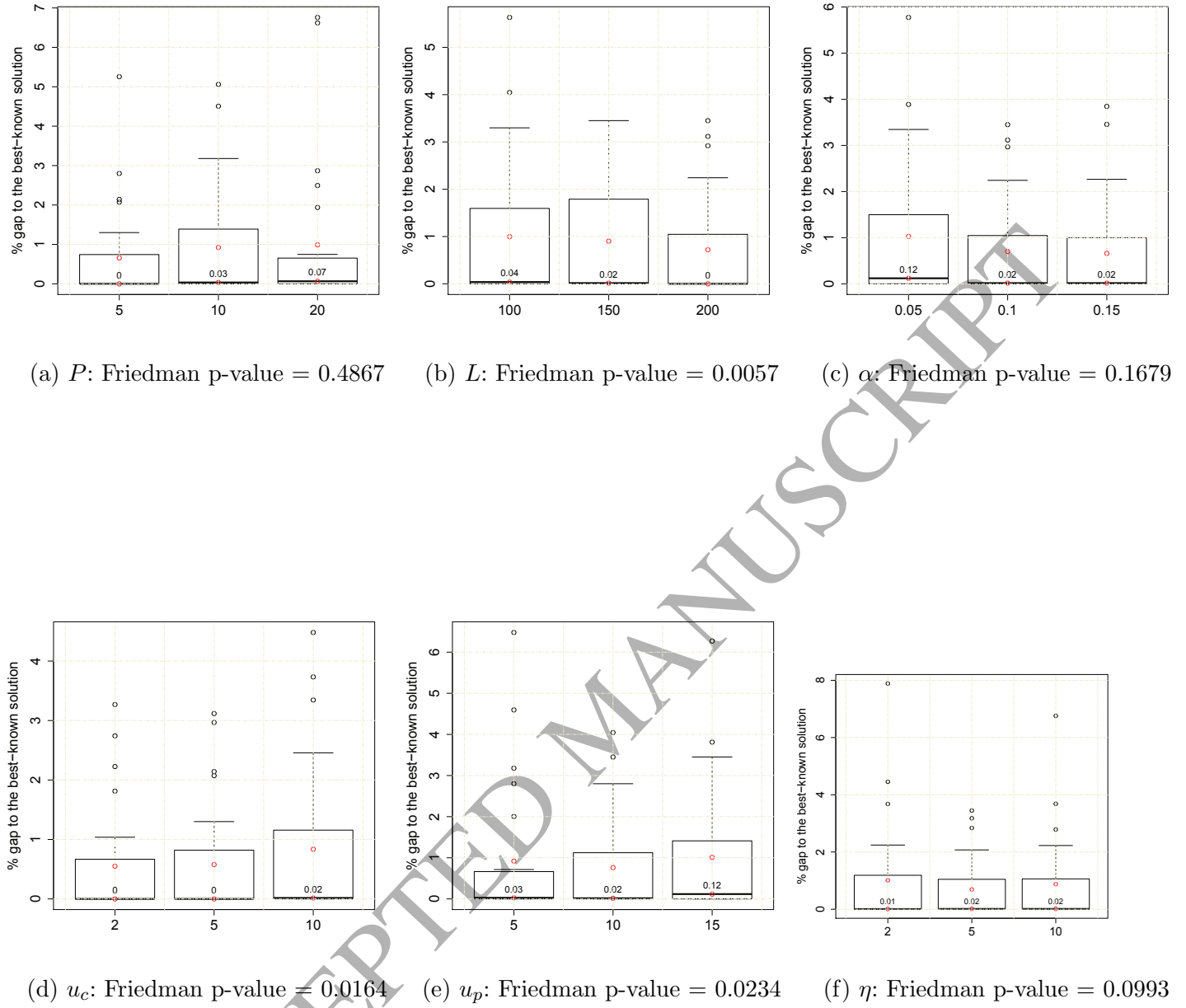


Fig. 5. Boxplots of the normalized average objective values for each considered value of a given parameter.

2, while fixing the other parameters to their final value (see Table 2). For this analysis, we use the same selection of 24 representative instances as for the parameter tuning with IRACE, and perform 30 independent runs for each considered value. The distribution and ranges of results are presented in the form of box-and-whisker plots (Figure 5). To determine whether there exists a statistically significant difference in solution samples for different values of a given parameter, we employ the Friedman rank sum test followed by a post-hoc analysis (i.e., Wilcoxon-signed rank tests) in cases when the Friedman test

reveals a deviation in performance.

The Friedman test shows a statistical difference in performance for L (p-value = 0.0057), u_c (p-value = 0.0164) and u_p (p-value = 0.0234), while the remaining parameters do not exhibit sensitivity. The results of the Wilcoxon-signed rank test are provided in Table 8.

Table 8

Wilcoxon-signed rank test

Parameter	Samples	p-value	Parameter	Samples	p-value	Parameter	Samples	p-value
L	100 vs. 150	0.3465	u_c	2 vs. 5	0.3280	u_p	5 vs. 10	0.2720
L	100 vs. 200	0.0076	u_c	2 vs. 10	0.0047	u_p	5 vs. 15	0.0961
L	150 vs. 200	0.0280	u_c	5 vs. 10	0.0186	u_p	10 vs. 15	0.0131

From the above results, we may conclude that the performance improves with higher values of L , which in turn requires more computing time with the increase of tabu search iterations. Similarly, the performance improves with lower values of u_c , which comes at the expense of higher computing time as the application frequency of the 3-opt operator increases.

6 Conclusion

The Orienteering Problem with Mandatory Visits and Exclusionary Constraints (OPMVEC) is an NP-hard problem derived from several practical situations. We present a highly effective memetic algorithm (MA) to tackle the problem. Extensive computational and statistical results conducted on six classes of 340 benchmark instances reveal that MA clearly outperforms the existing hybrid VNS algorithm, as well as the existing exact methods from the literature. Furthermore, the combined use of MA with integer programming, where MA is used to obtain an initial solution for CPLEX, leads to optimality for 87% of the benchmark instances. The proposed hybrid is able to prove for the first time optimality for 22 out of the 340 instances, which constitutes a strong motivation for its use.

For future work, new crossover operators could be integrated within the memetic framework, and other neighborhood operators could be considered for local improvement. Furthermore, a study of different OPMVEC extensions, as well as the design of more difficult instances, is very valuable.

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Appendix

For the purpose of future comparisons, Tables 9 to 14 give detailed results for the six classes of benchmark instances. The first two columns indicate the instance name and the number of nodes. The third column shows the objective value obtained with the OPMVEC-GG model [23] executed under CPLEX 12.6, followed by the required computing time in seconds. The fifth and the sixth column display the results reported with the proposed MA+OPMVEC-GG hybrid. An asterisk * next to an objective value denotes an optimal solution.

Columns 7-11 present detailed computational results for MA: the best objective value (f_{best}) over 30 independent runs, the percentage gap (gap_{best}) between the best objective values obtained with MA and VNS, the average objective value (f_{avg}) over 30 independent runs, the percentage gap (gap_{avg}) between the average MA and VNS objective values, and the average computing time in seconds. More precisely, the percentage gap between the two algorithms is defined as:

$$gap = 100 \times \frac{f_{MA} - f_{VNS}}{f_{MA}}.$$

The positive gap indicates that MA outperforms VNS, and is outperformed by VNS otherwise.

Columns 12-14 show detailed computational results for the hybrid VNS: the best objective value (f_{best}) over 30 independent runs, the average objective value (f_{avg}) and the average computing time in seconds. The entries in bold indicate the cases with improved performance.

Table 9
Results for instances from Class 1.

Instance	N	OPMVEC-GG		MA+OPMVEC-GG		MA					Hybrid VNS		
		<i>f</i>	<i>time(s)</i>	<i>f</i>	<i>time(s)</i>	<i>f_{best}</i>	<i>gap_{best}</i>	<i>f_{avg}</i>	<i>gap_{avg}</i>	<i>time(s)</i>	<i>f_{best}</i>	<i>f_{avg}</i>	<i>time(s)</i>
att48A	48	15*	15.7	15*	8.8	15	0.00	15.0	0.00	0.3	15	15.0	0.8
att48B	48	17*	13.4	17*	0.1	17	0.00	17.0	0.00	0.2	17	17.0	0.5
att48C	48	9*	33.7	9*	18.9	9	0.00	9.0	8.89	0.2	9	8.2	0.3
att48D	48	13*	5.3	13*	5.4	13	0.00	13.0	0.77	0.4	13	12.9	0.6
att48E	48	14*	16.4	14*	6.1	14	0.00	14.0	0.00	0.4	14	14.0	0.8
cmt121A	121	-	>3600	535*	1.3	535	0.00	535.0	0.00	4.3	535	535.0	3.3
cmt121B	121	-	>3600	416	>3600	416	0.00	416.0	0.10	4.2	416	415.6	10.1
cmt121C	121	476	>3600	501	>3600	501	0.00	500.0	0.10	5.4	501	499.5	8.7
cmt121D	121	-	>3600	530*	1972.9	530	0.00	530.0	0.00	5.5	530	530.0	3.9
cmt151A	151	815*	1433.0	815*	786.0	815	0.00	814.8	0.38	7.3	815	811.7	18.5
cmt151B	151	872*	3396.9	872*	4.2	872	0.00	872.0	0.00	5.9	872	872.0	4.9
cmt151C	151	-	>3600	435*	907.6	435	0.23	435.0	0.55	4.2	434	432.6	15.3
cmt151D	151	-	>3600	571*	1195.9	571	0.53	569.2	1.69	5.7	568	559.6	25.2
cmt151E	151	-	>3600	673*	956.2	673	0.00	672.6	0.79	5.1	673	667.3	21.2
cmt200A	200	-	>3600	598	>3600	598	0.17	597.1	0.87	12.3	597	591.9	28.0
cmt200B	200	-	>3600	1352*	10.4	1352	0.00	1352.0	0.00	42.8	1352	1352.0	13.8
cmt200C	200	-	>3600	751	>3600	751	0.13	750.1	0.20	16.1	750	748.6	44.7
cmt200D	200	-	>3600	919	>3600	919	1.96	915.7	2.05	21.5	901	896.9	46.9
cmt200E	200	-	>3600	1028*	3548.2	1028	1.26	1024.9	1.41	26.7	1015	1010.4	57.4
eil30A	30	6375*	2.8	6375*	0.0	6375	0.00	6375.0	0.00	0.1	6375	6375.0	0.2
eil30B	30	5125*	10.8	5125*	3.7	5125	0.00	5125.0	0.00	0.1	5125	5125.0	0.2
eil30C	30	5775*	8.5	5775*	2.7	5775	0.00	5775.0	0.00	0.1	5775	5775.0	0.2
eil30D	30	6275*	3.8	6275*	1.1	6275	0.00	6275.0	0.00	0.1	6275	6275.0	0.3
eil33A	33	5230*	88.8	5230*	13.0	5230	0.00	5230.0	0.00	0.1	5230	5230.0	0.3
eil33B	33	14380*	1.2	14380*	0.5	14380	0.00	14380.0	0.00	0.1	14380	14380.0	0.3
eil33C	33	7430*	34.5	7430*	10.4	7430	0.00	7430.0	0.00	0.1	7430	7430.0	0.4
eil33D	33	11630*	5.6	11630*	2.1	11630	0.00	11630.0	0.00	0.1	11630	11630.0	0.4
eil33E	33	12830*	6.1	12830*	0.8	12830	0.00	12830.0	0.00	0.1	12830	12830.0	0.4
eil51A	51	245*	7.3	245*	3.9	245	0.00	245.0	0.00	0.3	245	245.0	1.6
eil51B	51	287*	0.8	287*	0.1	287	0.00	287.0	0.00	0.3	287	287.0	0.6
eil51C	51	122*	35.5	122*	7.8	122	9.02	122.0	9.02	0.2	111	111.0	0.4
eil51D	51	150*	27.5	150*	9.8	150	0.00	150.0	1.40	0.2	150	147.9	0.9
eil51E	51	177*	32.2	177*	6.9	177	0.00	177.0	3.62	0.2	177	170.6	1.1
eil76A	76	520*	93.0	520*	22.4	520	0.00	519.4	0.39	1.2	520	517.4	3.8
eil76B	76	599*	48.2	599*	0.4	599	0.00	599.0	0.00	1.5	599	599.0	1.1
eil76C	76	232*	60.4	232*	25.1	232	0.00	232.0	0.52	0.6	232	230.8	1.8
eil76D	76	312*	28.0	312*	18.3	312	0.00	312.0	0.45	0.7	312	310.6	2.4
eil76E	76	367*	50.3	367*	20.2	367	0.00	367.0	0.08	0.8	367	366.7	2.8
eil101A	101	570*	2608.0	570*	125.4	570	0.88	567.3	1.18	2.5	565	560.6	8.7
eil101B	101	612*	5.2	612*	0.9	612	0.00	612.0	0.00	2.5	612	612.0	2.1
eil101C	101	-	>3600	281*	209.3	281	0.00	281.0	0.00	1.2	281	281.0	2.9
eil101D	101	367*	796.1	367*	140.5	367	0.00	367.0	0.00	1.5	367	367.0	4.2
eil101E	101	414*	438.9	414*	147.3	414	1.21	411.6	0.87	1.6	409	408.0	6.8
gil262A	262	-	>3600	4412	>3600	4412	1.77	4389.1	2.75	70.8	4334	4268.3	75.3
gil262B	262	-	>3600	4980*	28.9	4980	0.00	4980.0	0.00	83.7	4980	4980.0	26.3
gil262C	262	-	>3600	2556	>3600	2556	0.63	2554.1	1.61	26.9	2540	2513.1	77.4
gil262D	262	-	>3600	3213	>3600	3213	1.21	3185.8	2.24	38.3	3174	3114.5	101.4
gil262E	262	-	>3600	3692	>3600	3689	1.49	3687.6	2.80	55.6	3634	3584.5	122.7
op21A	21	165*	0.5	165*	0.3	165	0.00	165.0	0.00	0.0	165	165.0	0.1
op21B	21	135*	1.5	135*	0.4	135	0.00	135.0	0.00	0.0	135	135.0	0.0
op21C	21	150*	0.9	150*	0.4	150	0.00	150.0	0.00	0.0	150	150.0	0.1
op21D	21	155*	1.3	155*	0.3	155	0.00	155.0	0.00	0.0	155	155.0	0.1
op32A	32	85*	1.9	85*	1.0	85	0.00	85.0	0.00	0.1	85	85.0	0.3
op32B	32	115*	1.1	115*	0.9	115	0.00	115.0	0.00	0.1	115	115.0	0.2
op32C	32	35*	6.2	35*	1.8	35	0.00	35.0	0.00	0.1	35	35.0	0.0
op32D	32	60*	3.1	60*	1.3	60	0.00	60.0	0.00	0.1	60	60.0	0.1
op32E	32	75*	2.6	75*	1.2	75	0.00	75.0	0.00	0.1	75	75.0	0.2
op33A	33	260*	1.0	260*	1.0	260	0.00	260.0	0.00	0.1	260	260.0	0.3
op33B	33	330*	1.2	330*	0.5	330	0.00	330.0	0.00	0.2	330	330.0	0.2
op33C	33	110*	7.5	110*	2.1	110	0.00	110.0	0.00	0.1	110	110.0	0.1
op33D	33	160*	1.7	160*	1.0	160	0.00	160.0	0.00	0.1	160	160.0	0.1
op33E	33	180*	2.2	180*	1.7	180	0.00	180.0	0.00	0.1	180	180.0	0.2

Table 10
Results for instances from Class 2.

Instance	N	OPMVEC-GG		MA+OPMVEC-GG		MA					Hybrid VNS		
		<i>f</i>	<i>time(s)</i>	<i>f</i>	<i>time(s)</i>	<i>f_{best}</i>	<i>gap_{best}</i>	<i>f_{avg}</i>	<i>gap_{avg}</i>	<i>time(s)</i>	<i>f_{best}</i>	<i>f_{avg}</i>	<i>time(s)</i>
att48A	48	12*	15.2	12*	0.1	12	0.00	12.0	0.00	0.1	12	12.0	0.2
att48B	48	11*	853.0	11*	259.7	11	0.00	11.0	0.00	0.1	11	11.0	0.3
att48C	48	12*	7.2	12*	0.1	12	0.00	12.0	0.00	0.1	12	12.0	0.2
att48D	48	12*	0.7	12*	0.1	12	0.00	12.0	0.00	0.3	12	12.0	0.2
cmt121A	121	-	>3600	330*	0.6	330	0.00	330.0	0.21	5.1	330	329.3	3.3
cmt121B	121	-	>3600	305	>3600	305	1.64	304.4	1.64	4.7	300	299.4	5.1
cmt121C	121	-	>3600	330*	0.6	330	0.00	330.0	0.39	4.9	330	328.7	3.0
cmt121D	121	-	>3600	330*	0.7	330	0.00	330.0	0.00	4.3	330	330.0	1.8
cmt151A	151	462*	657.9	462*	196.3	462	0.43	458.8	1.20	6.1	460	453.3	7.5
cmt151B	151	-	>3600	561*	1.3	561	0.00	561.0	0.00	6.5	561	561.0	2.7
cmt151C	151	433*	374.7	433*	194.8	433	0.46	430.0	0.56	6.2	431	427.6	8.2
cmt151D	151	-	>3600	496*	167.1	496	0.00	495.1	1.11	7.4	496	489.6	9.6
cmt151E	151	541*	899.5	541*	147.6	541	0.00	538.2	0.20	7.3	541	537.1	6.9
cmt200A	200	-	>3600	825*	4.2	825	0.00	825.0	0.00	42.7	825	825.0	8.5
cmt200B	200	-	>3600	486*	741.2	482	0.21	474.6	2.93	22.6	481	460.7	20.2
cmt200C	200	-	>3600	625*	2067.2	625	2.40	616.3	2.60	33.4	610	600.3	26.4
cmt200D	200	-	>3600	732*	939.4	732	3.28	719.6	3.24	35.3	708	696.3	28.8
eil30A	30	2925*	1.7	2925*	0.8	2925	0.00	2925.0	0.00	0.0	2925	2925.0	0.1
eil30B	30	1625*	1.8	1625*	1.0	1625	0.00	1625.0	0.00	0.1	1625	1625.0	0.1
eil30C	30	1625*	2.4	1625*	1.3	1625	0.00	1625.0	0.00	0.0	1625	1625.0	0.0
eil30D	30	2475*	1.3	2475*	1.0	2475	0.00	2475.0	0.00	0.0	2475	2475.0	0.1
eil33A	33	8480*	1.2	8480*	0.3	8480	0.00	8480.0	0.00	0.1	8480	8480.0	0.1
eil33B	33	8480*	3.4	8480*	0.2	8480	0.00	8480.0	0.00	0.0	8480	8480.0	0.1
eil33C	33	8480*	2.2	8480*	1.1	8480	0.00	8480.0	0.00	0.0	8480	8480.0	0.1
eil33D	33	9430*	0.8	9430*	0.1	9430	0.00	9430.0	0.00	0.1	9430	9430.0	0.2
eil51A	51	168*	0.5	168*	0.1	168	0.00	168.0	0.00	0.1	168	168.0	0.3
eil51B	51	148*	24.1	148*	4.7	148	0.00	148.0	0.00	0.2	148	148.0	0.5
eil51C	51	168*	5.3	168*	0.1	168	0.00	168.0	0.00	0.3	168	168.0	0.4
eil51D	51	168*	0.6	168*	0.1	168	0.00	168.0	0.00	0.3	168	168.0	0.3
eil76A	76	377*	8.4	377*	11.6	377	0.00	377.0	0.00	0.8	377	377.0	1.0
eil76B	76	409*	1.7	409*	0.2	409	0.00	409.0	0.00	1.3	409	409.0	0.6
eil76C	76	201*	20.3	201*	10.4	201	0.00	196.9	-2.08	0.6	201	201.0	1.1
eil76D	76	294*	11.5	294*	11.7	294	0.34	293.7	0.24	0.9	293	293.0	1.1
eil76E	76	345*	7.3	345*	9.1	345	0.00	345.0	1.22	0.8	345	340.8	1.3
eil101A	101	386*	135.3	386*	62.2	386	0.78	383.9	1.17	2.6	383	379.4	3.8
eil101B	101	409*	54.4	409*	0.5	409	0.00	409.0	0.00	2.5	409	409.0	1.3
eil101C	101	260*	6.9	260*	254.5	260	0.00	260.0	1.50	1.8	260	256.1	2.4
eil101D	101	326*	55.3	326*	50.7	326	0.00	326.0	1.90	2.3	326	319.8	4.3
eil101E	101	370*	75.4	370*	64.3	370	0.00	370.0	1.08	2.7	370	366.0	4.0
gil262A	262	-	>3600	3003*	8.4	3003	0.00	3003.0	0.00	84.3	3003	3003.0	12.3
gil262B	262	-	>3600	2373	>3600	2373	3.08	2300.5	2.53	66.7	2300	2242.3	42.5
gil262C	262	-	>3600	2852	>3600	2852	1.72	2810.8	1.38	83.5	2803	2771.9	41.7
gil262D	262	-	>3600	3000*	2030.9	3000	0.63	2997.8	0.90	98.2	2981	2970.9	34.3
op21A	21	60*	0.4	60*	0.5	60	0.00	60.0	0.00	0.0	60	60.0	0.0
op21B	21	125*	0.3	125*	0.5	125	0.00	125.0	0.00	0.0	125	125.0	0.0
op21C	21	60*	0.4	60*	0.1	60	0.00	60.0	0.00	0.0	60	60.0	0.0
op21D	21	75*	2.8	75*	0.2	75	20.00	75.0	20.00	0.0	60	60.0	0.0
op21E	21	80*	3.1	80*	0.3	80	0.00	80.0	0.00	0.0	80	80.0	0.0
op32A	32	80*	1.9	80*	0.6	80	0.00	80.0	0.00	0.0	80	80.0	0.1
op32B	32	70*	1.8	70*	0.8	70	0.00	70.0	0.00	0.1	70	70.0	0.1
op32C	32	80*	0.3	80*	0.2	80	0.00	80.0	0.00	0.0	80	80.0	0.1
op33A	33	210*	0.5	210*	0.4	210	0.00	210.0	0.00	0.0	210	210.0	0.1
op33B	33	170*	4.3	170*	1.8	170	0.00	170.0	0.00	0.1	170	170.0	0.1
op33C	33	210*	0.9	210*	0.3	210	0.00	210.0	0.00	0.0	210	210.0	0.1
op33D	33	220*	2.3	220*	0.4	220	0.00	220.0	0.00	0.0	220	220.0	0.1

Table 11
Results for instances from Class 3.

Instance	N	OPMVEC-GG		MA+OPMVEC-GG		MA					Hybrid VNS		
		<i>f</i>	<i>time(s)</i>	<i>f</i>	<i>time(s)</i>	<i>f_{best}</i>	<i>gap_{best}</i>	<i>f_{avg}</i>	<i>gap_{avg}</i>	<i>time(s)</i>	<i>f_{best}</i>	<i>f_{avg}</i>	<i>time(s)</i>
att48A	48	4*	3.9	4*	1.3	4	0.00	4.0	0.00	0.2	4	4.0	0.2
att48B	48	7*	1.1	7*	0.1	7	0.00	7.0	0.00	0.1	7	7.0	0.2
att48C	48	4*	6.5	4*	24.8	4	0.00	4.0	0.00	0.1	4	4.0	0.2
att48D	48	5*	8.4	5*	9.9	5	0.00	5.0	0.00	0.1	5	5.0	0.2
att48E	48	5*	8.0	5*	3.7	5	0.00	5.0	0.00	0.6	5	5.0	0.2
cmt121A	121	-	>3600	152*	0.3	152	0.00	152.0	0.00	3.1	152	152.0	1.0
cmt121B	121	152*	67.6	152*	0.3	152	0.00	152.0	0.00	4.3	152	152.0	1.0
cmt121C	121	-	>3600	152*	0.4	152	0.00	152.0	0.00	1.1	152	152.0	1.0
cmt121D	121	152*	64.9	152*	0.3	152	0.00	152.0	0.00	0.6	152	152.0	1.0
cmt151A	151	-	>3600	199*	0.5	199	0.00	199.0	0.00	5.8	199	199.0	1.4
cmt151B	151	-	>3600	179*	52.2	179	0.00	178.2	2.69	5.3	179	173.4	2.2
cmt151C	151	-	>3600	199*	0.5	199	0.00	199.0	0.00	6.0	199	199.0	1.8
cmt151D	151	-	>3600	199*	0.5	199	0.00	199.0	0.00	2.7	199	199.0	1.4
cmt200A	200	-	>3600	463*	1.3	463	0.00	463.0	0.00	2.7	463	463.0	4.0
cmt200B	200	391*	370.5	391*	319.8	391	6.91	387.9	8.09	35.5	364	356.5	6.5
cmt200C	200	-	>3600	452*	291.8	452	3.10	449.8	3.36	54.3	438	434.7	6.8
cmt200D	200	-	>3600	463*	1.4	463	0.00	463.0	0.00	49.8	463	463.0	5.2
eil30A	30	950*	5.7	950*	2.0	950	0.00	950.0	0.00	0.1	950	950.0	0.0
eil30B	30	1950*	5.3	1950*	0.6	1950	0.00	1950.0	0.00	0.4	1950	1950.0	0.1
eil30C	30	2075*	0.1	2075*	0.1	2075	0.00	2075.0	0.00	0.0	2075	2075.0	0.1
eil30D	30	2075*	0.1	2075*	0.1	2075	0.00	2075.0	0.00	0.0	2075	2075.0	0.0
eil33A	33	5530*	0.4	5530*	0.1	5530	0.00	5530.0	0.00	0.0	5530	5530.0	0.1
eil33B	33	5530*	0.7	5530*	0.1	5530	0.00	5530.0	0.00	0.0	5530	5530.0	0.0
eil33C	33	5530*	0.2	5530*	0.1	5530	0.00	5530.0	0.00	0.0	5530	5530.0	0.0
eil51A	51	99*	1.4	99*	0.1	99	0.00	99.0	0.00	0.1	99	99.0	0.2
eil51B	51	90*	10.3	90*	3.5	90	0.00	90.0	0.00	0.1	90	90.0	0.2
eil51C	51	99*	5.1	99*	0.0	99	0.00	99.0	0.00	0.1	99	99.0	0.2
eil51D	51	99*	0.5	99*	0.0	99	0.00	99.0	0.00	0.3	99	99.0	0.1
eil76A	76	167*	5.5	167*	7.0	167	0.00	167.0	0.00	0.1	167	167.0	0.5
eil76B	76	202*	0.8	202*	0.1	202	0.00	202.0	0.00	1.1	202	202.0	0.3
eil76C	76	142*	4.7	142*	6.2	142	0.00	137.1	-3.57	0.9	142	142.0	0.5
eil76D	76	193*	23.6	193*	9.2	193	0.00	193.0	0.05	1.1	193	192.9	0.6
eil76E	76	202*	47.2	202*	0.1	202	0.00	202.0	0.00	1.2	202	202.0	0.4
eil101A	101	175*	34.6	175*	0.2	175	0.00	175.0	0.00	1.8	175	175.0	0.5
eil101B	101	155*	11.6	155*	17.5	155	0.00	139.7	-7.30	2.0	155	149.9	0.8
eil101C	101	175*	2416.5	175*	0.2	175	0.00	175.0	0.00	2.4	175	175.0	0.9
eil101D	101	175*	42.7	175*	0.1	175	0.00	175.0	0.00	2.6	175	175.0	0.5
gil262A	262	-	>3600	1369*	3.0	1369	0.00	1369.0	0.00	91.2	1369	1369.0	6.8
gil262B	262	-	>3600	1220	>3600	1220	5.49	1201.1	8.62	69.5	1153	1097.6	9.9
gil262C	262	-	>3600	1369*	2.9	1369	0.22	1368.5	0.53	84.7	1366	1361.3	12.7
gil262D	262	-	>3600	1369*	3.1	1369	0.00	1369.0	0.00	88.6	1369	1369.0	7.1
op21A	21	45*	0.0	45*	0.1	45	0.00	45.0	0.00	0.0	45	45.0	0.0
op21B	21	30*	0.1	30*	0.1	30	0.00	30.0	0.00	0.0	30	30.0	0.0
op21C	21	45*	0.0	45*	0.1	45	0.00	45.0	0.00	0.0	45	45.0	0.0
op21D	21	45*	0.0	45*	0.1	45	0.00	45.0	0.00	0.0	45	45.0	0.0
op32A	32	45*	0.5	45*	0.2	45	0.00	45.0	0.00	0.0	45	45.0	0.1
op32B	32	45*	1.0	45*	0.5	45	0.00	45.0	0.00	0.0	45	45.0	0.0
op32C	32	45*	0.3	45*	0.3	45	0.00	45.0	0.00	0.0	45	45.0	0.1
op32D	32	45*	0.1	45*	0.2	45	0.00	45.0	0.00	0.0	45	45.0	0.1
op33A	33	130*	0.1	130*	0.3	130	0.00	130.0	0.00	0.0	130	130.0	0.0
op33B	33	80*	0.2	80*	1.0	80	0.00	80.0	0.00	0.0	80	80.0	0.0
op33C	33	120*	0.3	120*	0.3	120	0.00	120.0	0.00	0.0	120	120.0	0.0
op33D	33	130*	0.8	130*	0.3	130	0.00	130.0	0.00	0.0	130	130.0	0.0

Table 12

Results for instances from Class 4.

Instance	N	OPMVEC-GG		MA+OPMVEC-GG		MA					Hybrid VNS		
		<i>f</i>	<i>time(s)</i>	<i>f</i>	<i>time(s)</i>	<i>f_{best}</i>	<i>gap_{best}</i>	<i>f_{avg}</i>	<i>gap_{avg}</i>	<i>time(s)</i>	<i>f_{best}</i>	<i>f_{avg}</i>	<i>time(s)</i>
att48A	48	21*	5.4	21*	5.3	21	0.00	21.0	3.33	0.9	21	20.3	2.1
att48B	48	29*	19.5	29*	9.4	29	3.45	29.0	4.48	2.4	28	27.7	2.7
att48C	48	13*	122.2	13*	11.6	13	7.69	13.0	8.46	0.4	12	11.9	0.7
att48D	48	17*	8.7	17*	7.0	17	0.00	17.0	1.18	0.6	17	16.8	1.4
att48E	48	20*	12.0	20*	4.9	20	0.00	20.0	5.00	0.7	20	19.0	1.8
cmt121A	121	-	>3600	814	>3600	814	0.61	811.2	0.57	12.1	809	806.6	39.8
cmt121B	121	-	>3600	513	>3600	513	0.19	511.0	0.06	8.0	512	510.7	33.6
cmt121C	121	-	>3600	669	>3600	669	0.60	662.9	1.16	13.5	665	655.2	51.9
cmt121D	121	-	>3600	774	>3600	774	1.03	772.4	1.73	16.7	766	759.0	47.2
cmt151A	151	1087*	2547.2	1087*	1073.9	1087	1.29	1082.8	2.75	24.5	1073	1053.0	74.7
cmt151B	151	1406	>3600	1483*	1126.2	1483	1.48	1476.4	1.59	49.7	1461	1452.9	53.9
cmt151C	151	-	>3600	477*	1653.7	477	0.21	476.6	0.86	7.2	476	472.5	22.5
cmt151D	151	-	>3600	647*	1362.1	647	0.62	645.5	2.12	9.9	643	631.8	35.0
cmt151E	151	-	>3600	799*	1340.6	799	2.50	799.0	3.28	14.1	779	772.8	53.1
cmt200A	200	-	>3600	723	>3600	723	0.00	713.6	-0.38	22.8	723	716.3	36.7
cmt200B	200	-	>3600	2171*	3409.2	2164	1.85	2149.3	1.69	128.3	2124	2112.9	195.7
cmt200C	200	-	>3600	908	>3600	908	0.77	905.1	2.12	34.8	901	885.9	97.5
cmt200D	200	-	>3600	1151	>3600	1151	2.43	1142.8	3.47	58.2	1123	1103.2	133.8
cmt200E	200	-	>3600	1324*	2117.6	1323	2.49	1313.2	3.13	42.6	1290	1272.1	185.3
eil30A	30	9375*	18.7	9375*	59.6	9375	0.00	9375.0	0.03	0.1	9375	9372.5	0.9
eil30B	30	5450*	60.4	5450*	46.7	5450	0.00	5450.0	0.00	0.1	5450	5450.0	0.4
eil30C	30	7300*	30.2	7300*	87.9	7300	0.00	7300.0	0.00	0.1	7300	7300.0	0.7
eil30D	30	8575*	31.4	8575*	54.6	8575	0.00	8575.0	0.00	0.4	8575	8575.0	0.9
eil33A	33	6950*	41.8	6950*	29.2	6950	0.00	6950.0	0.00	0.4	6950	6950.0	0.5
eil33B	33	19730*	9.2	19730*	16.7	19730	0.00	19730.0	0.00	0.4	19730	19730.0	1.0
eil33C	33	10630*	77.4	10630*	44.1	10630	0.00	10630.0	0.00	0.4	10630	10630.0	0.9
eil33D	33	14180*	9.6	14180*	10.6	14180	0.00	14180.0	0.00	0.4	14180	14180.0	0.9
eil33E	33	17430*	7.9	17430*	4.6	17430	0.00	17430.0	0.00	0.5	17430	17430.0	1.1
eil51A	51	328*	45.7	328*	41.7	328	0.91	328.0	1.19	0.7	325	324.1	2.7
eil51B	51	509*	21.8	509*	15.5	509	0.00	507.2	0.22	1.8	509	506.1	2.6
eil51C	51	168*	11.5	168*	27.4	168	0.00	168.0	0.00	0.2	168	168.0	0.7
eil51D	51	201*	39.2	201*	11.6	201	1.49	200.7	1.35	0.3	198	198.0	1.2
eil51E	51	232*	28.3	232*	13.9	232	0.00	232.0	0.00	0.4	232	232.0	1.9
eil76A	76	629*	103.6	629*	93.2	629	0.00	629.0	0.78	3.3	629	624.1	12.2
eil76B	76	923*	343.8	923*	153.5	923	0.87	922.1	1.21	7.6	915	910.9	9.4
eil76C	76	274*	76.5	274*	117.4	274	0.00	274.0	0.00	1.0	274	274.0	3.5
eil76D	76	366*	73.1	366*	54.2	366	0.00	366.0	0.27	1.1	366	365.0	5.6
eil76E	76	442*	20.4	442*	67.6	442	0.00	442.0	0.43	1.4	442	440.1	7.7
eil101A	101	789*	781.7	789*	535.3	789	1.39	787.0	2.05	6.6	778	770.9	29.8
eil101B	101	1071*	1244.0	1071*	345.8	1071	0.93	1068.4	1.09	15.8	1061	1056.8	18.4
eil101C	101	326*	660.6	326*	324.1	326	0.00	326.0	0.00	1.4	326	326.0	4.4
eil101D	101	422*	342.7	422*	472.8	422	0.00	422.0	0.00	2.2	422	422.0	9.6
eil101E	101	491*	842.2	491*	429.7	491	0.00	491.0	0.24	3.1	491	489.8	11.2
gil262A	262	-	>3600	5642	>3600	5642	3.83	5578.4	4.13	154.7	5426	5347.8	370.9
gil262B	262	-	>3600	7992	>3600	7988	2.42	7947.8	3.02	427.6	7795	7707.7	362.2
gil262C	262	-	>3600	2997	>3600	2997	0.47	2986.6	1.81	47.4	2983	2932.4	125.3
gil262D	262	-	>3600	3883	>3600	3883	5.49	3816.6	4.83	75.2	3670	3632.4	206.6
gil262E	262	-	>3600	4413	>3600	4413	2.52	4371.7	3.31	106.9	4302	4227.2	245.7
op21A	21	260*	4.7	260*	1.4	260	0.00	260.0	0.00	0.1	260	260.0	0.3
op21B	21	180*	8.0	180*	1.2	180	0.00	180.0	0.00	0.1	180	180.0	0.1
op21C	21	215*	2.8	215*	1.5	215	0.00	215.0	0.00	0.1	215	215.0	0.3
op21D	21	260*	2.7	260*	1.1	260	0.00	260.0	0.00	0.1	260	260.0	0.3
op32A	32	110*	1.3	110*	0.6	110	0.00	110.0	0.00	0.2	110	110.0	0.6
op32B	32	180*	5.7	180*	1.1	180	0.00	180.0	0.00	0.1	180	180.0	0.4
op32C	32	35*	10.8	35*	2.8	35	0.00	35.0	0.00	0.1	35	35.0	0.1
op32D	32	75*	1.0	75*	1.0	75	0.00	75.0	0.00	0.1	75	75.0	0.3
op32E	32	90*	2.8	90*	1.5	90	0.00	90.0	0.00	0.1	90	90.0	0.4
op33A	33	310*	4.1	310*	2.0	310	0.00	310.0	0.00	0.2	310	310.0	0.7
op33B	33	460*	3.4	460*	2.0	460	0.00	460.0	0.00	0.5	460	460.0	0.5
op33C	33	120*	4.1	120*	1.4	120	0.00	120.0	0.00	0.1	120	120.0	0.1
op33D	33	180*	5.7	180*	2.7	180	0.00	180.0	0.00	0.1	180	180.0	0.2
op33E	33	220*	2.4	220*	1.5	220	0.00	220.0	0.00	0.1	220	220.0	0.3

Table 13
Results for instances from Class 5.

Instance	N	OPMVEC-GG		MA+OPMVEC-GG		MA					Hybrid VNS		
		<i>f</i>	<i>time(s)</i>	<i>f</i>	<i>time(s)</i>	<i>f_{best}</i>	<i>gap_{best}</i>	<i>f_{avg}</i>	<i>gap_{avg}</i>	<i>time(s)</i>	<i>f_{best}</i>	<i>f_{avg}</i>	<i>time(s)</i>
att48A	48	24*	21.7	24*	6.8	24	0.00	24.0	0.00	1.7	24	24.0	2.2
att48B	48	19*	21.3	19*	7.2	19	0.00	19.0	0.53	0.6	19	18.9	1.4
att48C	48	23*	25.0	23*	5.1	23	0.00	23.0	0.43	1.3	23	22.9	1.8
att48D	48	25*	21.7	25*	7.8	25	0.00	25.0	0.00	1.6	25	25.0	1.9
cmt121A	121	-	>3600	640	>3600	640	0.47	639.8	2.30	15.0	637	625.1	27.8
cmt121B	121	-	>3600	482	>3600	482	3.53	475.4	3.39	13.7	465	459.3	32.6
cmt121C	121	-	>3600	636	>3600	636	1.57	635.0	2.82	15.3	626	617.1	26.1
cmt121D	121	-	>3600	710	>3600	710	1.69	708.0	2.19	9.8	698	692.5	25.3
cmt151A	151	681*	2010.5	681*	1039.7	681	0.44	655.8	-0.32	20.4	678	657.9	27.2
cmt151B	151	-	>3600	1207*	617.5	1207	0.91	1204.9	1.73	48.7	1196	1184.0	32.7
cmt151C	151	610*	988.2	610*	1045.5	610	0.66	563.4	-5.41	18.9	606	593.9	21.4
cmt151D	151	-	>3600	773*	1096.5	773	2.85	754.6	1.26	23.8	751	745.1	38.5
cmt151E	151	914*	829.2	914*	692.7	914	3.83	902.7	3.90	32.6	879	867.5	45.5
cmt200A	200	-	>3600	1678*	1403.1	1665	1.32	1653.3	1.85	235.4	1643	1622.7	136.5
cmt200B	200	-	>3600	760*	1348.0	760	7.37	731.3	8.27	56.1	704	670.8	46.0
cmt200C	200	1053*	2617.9	1053*	1316.2	1053	6.36	1047.0	9.54	119.7	986	947.1	106.8
cmt200D	200	-	>3600	1262*	1186.8	1261	5.47	1242.4	6.25	54.6	1192	1164.8	192.6
eil30A	30	6225*	8.6	6225*	11.6	6225	0.00	6225.0	0.00	0.3	6225	6225.0	0.6
eil30B	30	3350*	8.5	3350*	9.7	3350	0.00	3350.0	0.00	0.2	3350	3350.0	0.3
eil30C	30	4750*	9.8	4750*	7.6	4750	0.00	4750.0	0.00	0.2	4750	4750.0	0.3
eil30D	30	5075*	15.7	5075*	11.6	5075	0.00	5075.0	0.00	0.3	5075	5075.0	0.5
eil33A	33	14230*	7.7	14230*	3.2	14230	0.00	14230.0	0.00	0.7	14230	14230.0	0.8
eil33B	33	12880*	8.6	12880*	3.7	12880	0.00	12880.0	0.06	0.4	12880	12872.0	0.7
eil33C	33	14930*	3.0	14930*	3.0	14930	0.00	14930.0	0.00	0.6	14930	14930.0	0.7
eil33D	33	15680*	19.1	15680*	4.6	15680	0.00	15680.0	0.00	0.5	15680	15680.0	0.7
eil51A	51	402*	3.6	402*	11.2	402	0.00	401.9	1.02	1.5	402	397.8	2.2
eil51B	51	236*	61.8	236*	8.7	236	0.42	236.0	0.42	0.9	235	235.0	1.5
eil51C	51	302*	74.2	302*	10.7	302	0.99	301.0	1.96	0.9	299	295.1	2.4
eil51D	51	357*	44.2	357*	8.5	357	0.00	355.9	1.66	1.3	357	350.0	3.0
eil76A	76	481*	51.2	481*	35.3	481	0.00	479.3	-0.13	2.7	481	479.9	6.8
eil76B	76	754*	77.7	754*	61.8	754	0.93	751.0	0.99	6.8	747	743.6	7.4
eil76C	76	228*	125.3	228*	42.3	228	0.00	224.0	1.65	1.0	228	220.3	2.7
eil76D	76	341*	94.5	341*	40.3	341	0.00	338.6	1.57	1.5	341	333.3	4.7
eil76E	76	411*	128.5	411*	29.0	411	0.24	408.2	1.13	2.1	410	403.6	5.6
eil101A	101	559*	374.0	559*	162.8	559	2.50	547.7	1.73	5.7	545	538.2	16.5
eil101B	101	857*	312.2	857*	173.9	857	0.00	856.8	1.09	16.7	857	847.5	14.5
eil101C	101	296*	159.8	296*	16.3	296	0.00	295.8	1.89	4.0	296	290.2	5.6
eil101D	101	418*	235.6	418*	156.5	418	1.67	411.0	1.39	4.6	411	405.3	14.1
eil101E	101	521*	104.8	521*	134.1	521	3.84	506.8	2.58	5.7	501	493.7	15.7
gil262A	262	-	>3600	6352	>3600	6352	1.59	6308.7	2.07	455.8	6251	6178.0	202.2
gil262B	262	-	>3600	3618	>3600	3618	1.71	3484.1	6.31	149.2	3556	3264.1	130.7
gil262C	262	-	>3600	4875	>3600	4861	3.31	4757.3	5.90	248.5	4700	4476.5	313.6
gil262D	262	-	>3600	5578	>3600	5578	4.88	5555.2	5.86	319.2	5306	5229.4	248.0
op21A	21	135*	0.4	135*	0.4	135	0.00	135.0	0.00	0.1	135	135.0	0.0
op21B	21	235*	0.2	235*	0.6	235	0.00	235.0	0.00	0.1	235	235.0	0.1
op21C	21	135*	0.5	135*	0.5	135	0.00	135.0	0.00	0.1	135	135.0	0.0
op21D	21	135*	0.7	135*	0.6	135	0.00	135.0	0.00	0.1	135	135.0	0.0
op21E	21	165*	1.4	165*	1.0	165	0.00	165.0	0.00	0.1	165	165.0	0.0
op32A	32	150*	1.5	150*	0.8	150	0.00	150.0	0.00	0.5	150	150.0	0.4
op32B	32	110*	2.4	110*	1.2	110	0.00	110.0	0.00	0.3	110	110.0	0.5
op32C	32	140*	1.2	140*	2.9	140	0.00	140.0	0.00	0.4	140	140.0	0.5
op33A	33	400*	1.9	400*	1.8	400	0.00	400.0	0.00	0.3	400	400.0	0.5
op33B	33	270*	15.3	270*	4.2	270	0.00	270.0	0.00	0.2	270	270.0	0.2
op33C	33	410*	0.8	410*	1.6	410	0.00	410.0	0.00	0.6	410	410.0	0.5
op33D	33	440*	1.9	440*	4.3	440	0.00	440.0	0.00	0.6	440	440.0	0.4

Table 14
Results for instances from Class 6.

Instance	N	OPMVEC-GG		MA+OPMVEC-GG		MA					Hybrid VNS		
		<i>f</i>	<i>time(s)</i>	<i>f</i>	<i>time(s)</i>	<i>f_{best}</i>	<i>gap_{best}</i>	<i>f_{avg}</i>	<i>gap_{avg}</i>	<i>time(s)</i>	<i>f_{best}</i>	<i>f_{avg}</i>	<i>time(s)</i>
att48A	48	11*	2.1	11*	3.9	11	0.00	11.0	0.00	0.9	11	11.0	0.7
att48B	48	19*	25.2	19*	8.5	19	5.26	19.0	5.26	1.6	18	18.0	1.5
att48C	48	12*	2.6	12*	2.0	12	0.00	12.0	0.00	0.7	12	12.0	0.8
att48D	48	14*	2.6	14*	45.9	14	0.00	14.0	0.00	1.2	14	14.0	1.2
att48E	48	16*	2.3	16*	2.0	16	6.25	16.0	6.25	0.8	15	15.0	1.4
cmt121A	121	-	>3600	503	>3600	503	0.80	502.8	2.19	23.8	499	491.8	29.4
cmt121B	121	-	>3600	458	>3600	458	0.22	457.9	1.92	21.3	457	449.1	28.5
cmt121C	121	-	>3600	557	>3600	557	0.54	556.9	1.58	23.4	554	548.1	24.6
cmt121D	121	-	>3600	618	>3600	618	0.97	617.8	1.29	33.1	612	609.8	11.4
cmt151A	151	-	>3600	873*	565.9	873	2.18	868.8	2.85	50.5	854	844.0	33.2
cmt151B	151	-	>3600	549*	502.9	549	3.10	542.5	3.96	26.7	532	521.0	31.4
cmt151C	151	740*	2341.1	740*	503.2	740	4.19	736.4	5.12	34.1	709	698.7	48.0
cmt151D	151	852*	551.0	852*	473.2	852	2.23	850.3	3.53	55.7	833	820.3	44.1
cmt200A	200	-	>3600	1445*	1450.4	1442	2.70	1431.7	3.81	239.1	1403	1377.2	100.8
cmt200B	200	-	>3600	955*	1177.2	926	3.02	910.7	6.39	125.8	898	852.5	59.5
cmt200C	200	1224	>3600	1224*	1015.0	1224	5.39	1206.7	7.47	179.4	1158	1116.6	98.3
cmt200D	200	1363	>3600	1386*	1048.9	1385	4.55	1373.5	5.21	186.7	1322	1301.9	103.0
eil30A	30	1550*	15.1	1550*	8.9	1550	0.00	1550.0	0.00	0.1	1550	1550.0	0.1
eil30B	30	4775*	12.8	4775*	16.3	4775	0.00	4775.0	0.00	0.1	4775	4775.0	0.3
eil30C	30	5550*	1.3	5550*	1.2	5550	0.00	5550.0	0.00	0.4	5550	5550.0	0.3
eil30D	30	5850*	7.2	5850*	0.1	5850	2.14	5787.5	1.08	0.1	5725	5725.0	0.1
eil33A	33	10600*	7.0	10600*	4.2	10600	0.00	10600.0	0.00	0.5	10600	10600.0	0.4
eil33B	33	13330*	6.5	13330*	1.9	13330	0.00	13330.0	0.00	0.4	13330	13330.0	0.4
eil33C	33	14030*	5.6	14030*	0.1	14030	0.00	14030.0	0.00	0.5	14030	14030.0	0.2
eil51A	51	329*	3.0	329*	7.5	329	0.00	328.6	0.18	1.3	329	328.0	1.8
eil51B	51	212*	6.4	212*	6.0	212	0.00	212.0	0.38	0.9	212	211.2	0.9
eil51C	51	289*	1.7	289*	7.0	289	0.00	288.2	0.62	1.2	289	286.4	1.7
eil51D	51	344*	12.5	344*	8.7	344	0.00	344.0	0.03	1.6	344	343.9	1.5
eil76A	76	233*	22.2	233*	21.2	233	0.00	233.0	0.52	2.3	233	231.8	2.0
eil76B	76	613*	161.2	613*	41.2	613	1.79	607.1	1.47	5.8	602	598.2	4.1
eil76C	76	211*	19.7	211*	22.7	211	0.00	211.0	0.05	1.8	211	210.9	2.0
eil76D	76	305	>3600	305*	31.7	305	0.33	304.0	0.20	2.7	304	303.4	3.7
eil76E	76	386*	247.3	386*	30.6	386	0.78	383.2	0.65	3.5	383	380.7	6.2
eil101A	101	633*	211.2	633*	236.1	633	1.58	628.8	1.32	14.8	623	620.5	9.5
eil101B	101	344*	363.7	344*	192.5	344	0.87	342.9	2.16	8.5	341	335.5	5.2
eil101C	101	474*	257.8	474*	215.6	474	1.48	464.0	0.63	8.6	467	461.1	13.5
eil101D	101	-	>3600	555*	160.7	555	1.44	552.8	1.74	12.8	547	543.2	14.4
gil262A	262	-	>3600	4780	>3600	4780	3.60	4700.5	3.45	430.2	4608	4538.1	177.4
gil262B	262	-	>3600	2927	>3600	2927	6.97	2647.8	4.55	198.7	2723	2527.2	45.3
gil262C	262	-	>3600	4070	>3600	4070	4.99	3940.6	4.31	337.6	3867	3770.8	191.9
gil262D	262	-	>3600	4665	>3600	4665	4.22	4583.9	4.25	447.3	4468	4388.9	167.3
op21A	21	155*	1.3	155*	0.5	155	0.00	155.0	0.00	0.1	155	155.0	0.1
op21B	21	75*	3.3	75*	0.4	75	0.00	75.0	0.00	0.1	75	75.0	0.0
op21C	21	110*	1.1	110*	1.7	110	0.00	110.0	0.00	0.1	110	110.0	0.1
op21D	21	160*	2.2	160*	2.5	160	0.00	160.0	0.00	0.1	160	160.0	0.1
op32A	32	115*	0.6	115*	0.8	115	0.00	115.0	0.00	0.1	115	115.0	0.4
op32B	32	65*	2.0	65*	3.4	65	0.00	65.0	0.00	0.2	65	65.0	0.1
op32C	32	105*	0.9	105*	1.2	105	0.00	105.0	0.00	0.4	105	105.0	0.4
op32D	32	130*	2.2	130*	1.1	130	0.00	130.0	0.00	0.2	130	130.0	0.3
op33A	33	350*	3.8	350*	1.0	350	0.00	350.0	0.00	0.1	350	350.0	0.3
op33B	33	130*	3.1	130*	2.8	130	0.00	130.0	0.00	0.1	130	130.0	0.1
op33C	33	240*	6.5	240*	3.2	240	0.00	240.0	0.00	0.3	240	240.0	0.3
op33D	33	300*	2.4	300*	1.5	300	0.00	300.0	0.00	0.1	300	300.0	0.3